



Beyond Accuracy: Stability-Oriented Learning for TCRI Credit Risk Prediction

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Outline

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2. Introduction
3. Data
4. Benchmark
5. Study plan
6. Results



Motivation

- ▣ Credit risk prediction is crucial for financial stability
- ▣ Taiwan's Standard: TCRI (Taiwan Corporate Credit Risk Index).
 - ▣ Covers 90% of listed companies
 - ▣ Key metric for lending decisions
- ▣ The Problem: Existing TCRI studies lack comprehensive predictive modeling



Let's start with the motivation. As we know, credit risk is the most critical issue for banks and investors. If we can accurately predict which company will fail, we can ensure financial stability.

In Taiwan, we have TCRI. It covers over 90% of publicly listed companies and is widely used by banks for lending decisions.

However, even though TCRI is so important, we found that few studies have built a complete machine learning system to predict it. This is where our story begins.

Introduction

- ▣ Past Approaches:
 - ▣ Financial Ratios (e.g., Altman Z-score).
 - ▣ Machine Learning (SVM, Random Forest, Neural Networks).
- ▣ What is Missing?
 - ▣ Temporal Leakage: Ignoring the time factor in data splits
 - ▣ Calibration: Probabilities are not "real" risk scores
 - ▣ Interpretability: "Black box" models are hard to trust



In the past, people used simple financial ratios, like the famous Altman Z-score.

Later, researchers started using Machine Learning, like SVM and Random Forests, which improved accuracy.

But, after reviewing the literature, we found three major gaps in prior work:

1. Temporal Leakage: Many studies ignore the 'time' factor. They shuffle data randomly, which is like using the future to predict the past. That is cheating.
2. Lack of Calibration: Most models give a score, but not a real 'probability.'
3. Interpretability: Advanced models are often 'black boxes.' Banks need to know why a company is risky, not just the result.

Introduction

- ▣ Key Methodologies:
 - ▣ Time-Aware Split: Prevent information leakage
 - ▣ Probabilistic Calibration: Interpretable risk scores (Platt Rescale)
 - ▣ Interpretability: SHAP values for feature importance
- ▣ Goal: A transparent, stable, and practical decision-support tool



To solve these problems, we propose a new framework using data from the TEJ database.

Our approach has three key features corresponding to the gaps I just mentioned:

1. Time-Aware Split: We strictly separate training and testing data by year to prevent leakage.
2. Calibration: We use techniques like Platt Rescaling to make sure our predicted probability is realistic.
3. Explainability: We use SHAP values to explain our model, making it transparent for decision-makers.

In short, our goal is to build a tool that is not only accurate but also stable and trustworthy for real-world use.

Data

- ▣ Dependent variable: TCRI score
- ▣ Independent variable: the most significant financial ratios identified in Beaver and Altman
 - ▶ Working Capital / Total Assets
 - ▶ Retained Earnings / Total Assets
 - ▶ Cash Flow / Total Debt
 - ▶ Total Debt / Total Assets
 - ▶ Current Ratio



Data Preparation and Cleaning

- ▣ To ensure data quality, we addressed missing values and removed extreme outliers before analysis.
- 1. Missing Value Imputation
 - ▶ Create a missing value indicator for each independent variables
 - ▶ imputed using the median of firms with the same first two digits of the company code.
 - ▶ If the whole group are missing → use the overall sample median
- 2. Outlier Removal
 - ▶ Observations where any ratio exceeded three standard deviations from the mean were excluded
- 3. Take the logarithm



To ensure data quality, we addressed missing values and removed extreme outliers before analysis and then take the logarithm

Key Descriptive Statistics

	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	3.13318	0	0.0964763	0	3.13318
mean	0.260084	0.111762	0.109787	0.457993	3.04299
std	0.205335	1.27573	0.49632	0.195864	11.1524
min	-0.742243	-82.0719	-18.1768	0.00303337	0.0588412
50%	0.251428	0.150736	0.080471	0.454693	1.79589
max	0.996967	1.58174	6.63415	1.05463	711.495
skewness	0.0955887	-44.8339	-10.2512	0.228632	29.3605
kurtosis	0.472559	2247.58	264.752	-0.219784	1243.48

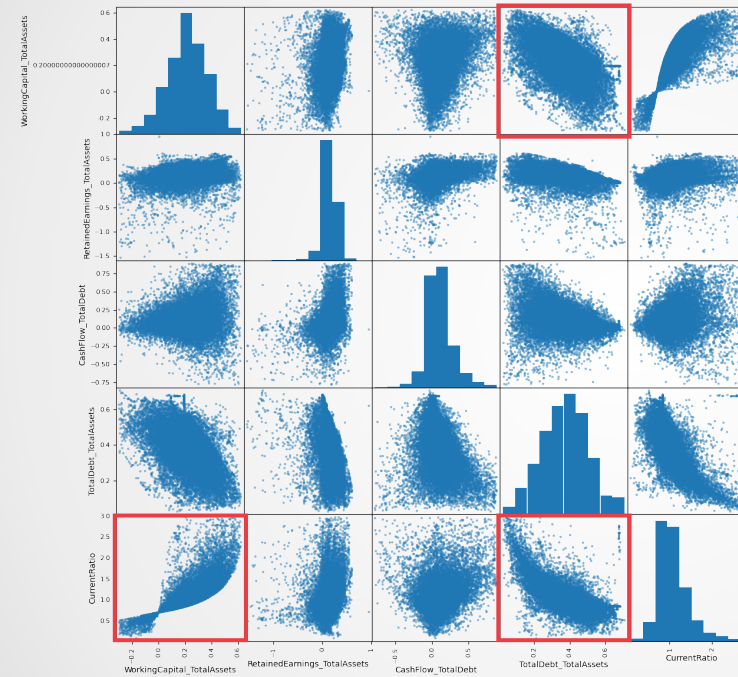


Data cleaning

	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	0	0	0	0	0
mean	0.215023	0.135149	0.103381	0.373306	1.08464
std	0.15288	0.167935	0.186678	0.129052	0.380631
min	-0.298188	-1.53474	-0.78052	0.0258126	0.0741276
50%	0.215281	0.141115	0.0783935	0.378056	1.00802
max	0.623079	0.948465	0.885164	0.706987	2.96362
skewness	-0.242851	-2.41152	0.543679	-0.0259596	1.34961
kurtosis	0.0358163	15.6336	2.59051	-0.345129	2.99375



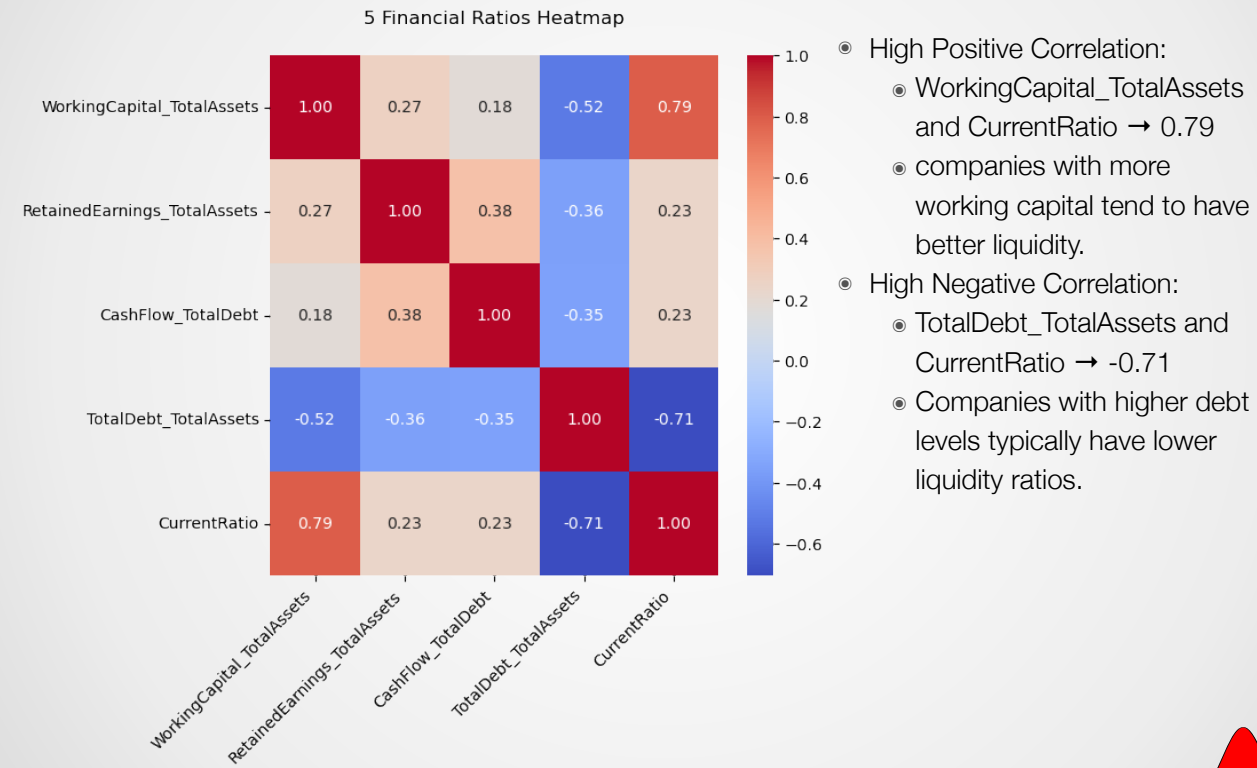
Data exploration



- Strong Linear Patterns:
 - TotalDebt_TotalAssets and WorkingCapital_TotalAssets (negative slopes).
- Non-linear Relationships:
 - CurrentRatio vs WorkingCapital_TotalAssets,
 - CurrentRatio vs TotalDebt_TotalAssets.

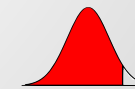


Data exploration



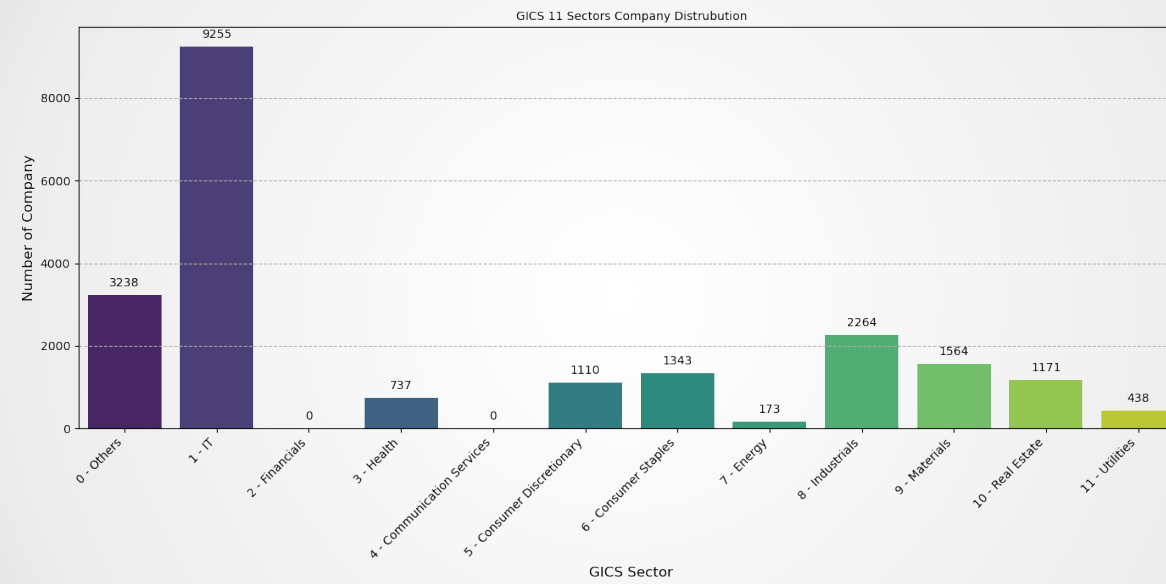
Data exploration

dummy variable	GICS Industry Sector	dummy variable	GICS Industry Sector
1	Information Technology	6	Consumer Staples
2	Financials	7	Energy
3	Health Care	8	Industrials
4	Communication Services	9	Materials
5	Consumer Discretionary	10	Real Estate
		11	Utilities



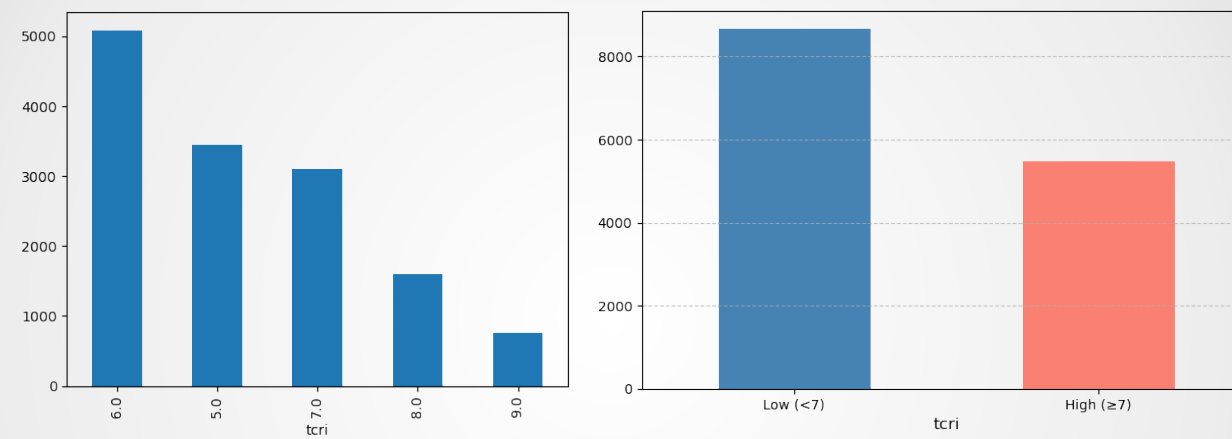
We also classify companies into eleven industries and others based on the Global Industry Classification Standard.

Data exploration

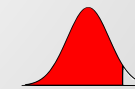


This is the Distribution of industry sector

Data exploration (TCRI)



- The grouped distribution is not severely imbalanced, though some moderate skew remains.



Data exploration

- Ratios provide **complementary signals**—moderate overlaps between leverage and liquidity, but largely **distinct information profiles**, supporting their joint use in **credit rating modeling** and **SHAP analysis**.



Benchmark

- ▣ Dataset (2014-2024):
 - ▣ TEJ financial ratio
 - ▣ TCRI ratings
- ▣ Use **t-year** financial ratios to predict **t+1** high-risk status (binary)
 - ▣ High risk (1): $\text{tcri} \geq 7$
 - ▣ Low risk (0): $\text{tcri} < 7$

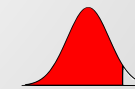


Benchmark

▣ Sample Size / Positive Rate

		n	pos	pos_rate_%
split				
2015–2022	Train	4624	1602	34.65
2023	Valid	616	221	35.88
2024	Test	619	218	35.22

▣ Logistic Regression



Our task is use year-t financial ratios to predict next-year high risk; the label is $TCRI \geq 7$ or D. We drop each firm's last year without a t+1 label to avoid leakage.

We split by forecast year:

Train 2014–2021,

Valid 2022,

Test 2023–2024.

Our benchmark model is Logistic Regression, followed by calibration and threshold selection on the validation set.

Test Performance

- ▣ Platt rescale
 - ▣ Probabilities more reliable
 - ▣ Operating-point trade-off
 - ▣ Precision
 - ▣ Recall
 - ▣ F1

	f1	precision	recall	brier
raw	0.7017	0.6125	0.8212	0.1556
platt	0.7043	0.6788	0.7318	0.1416



we use platt calibrated on test to make the probabilities more reliable

Platt transform raw scores into well-calibrated probabilities to make the f1 better

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Brier score

Definition: the mean squared error of probabilistic predictions,

$$1/n[\sigma(\pi - y_i)^2]$$

. Lower is better.

Interpretation here: Going from 0.1555 to 0.1415 (-9%) means your predicted probabilities are better calibrated—"a 30% score actually occurs ~30% of the time" more often after Platt.

Precision

Definition:

T

P

/

(

T

P

+

F

P

)

TP/(TP+FP). Of the cases you flagged as high risk, how many were truly high risk.

Why it increased: You're using the same threshold (t_{f1} picked on Valid). After Platt rescales scores, more borderline negatives move below that threshold, so FP drops, precision rises.

Recall

Definition:

T

P

/

(

T

P

+

F

N

)

TP/(TP+FN). Of all true high-risk cases, how many you caught.

Why it decreased: The same rescaling also pushes some borderline positives below the fixed threshold, so FN rises, recall falls.

F1

Definition: harmonic mean of precision and recall.

Why it nudged up: The precision gain slightly outweighs the recall loss at this operating point, so F1 rises from 0.7017 to 0.7043.

t_f1 is 0.466 which will have highest f1

Forecast Result

- ▣ Captures **~82%** of truly high-risk
- ▣ About **60%** flagged firms are truly high-risk

	n	tp	fp	fn	tn	precision	recall	accuracy
forecast_year								
2024	619	147	93	32	347	0.6125	0.821229	0.798061



This page summarizes the forecast year 2024 result, we use 2023 financial ratios to predict whether 2024 is high risk or not.

The operating threshold is $t_{f1} = 0.466$ chosen on the validation set, and probabilities are Platt-calibrated.

The test set contains 619 firms; the true high-risk count is 179

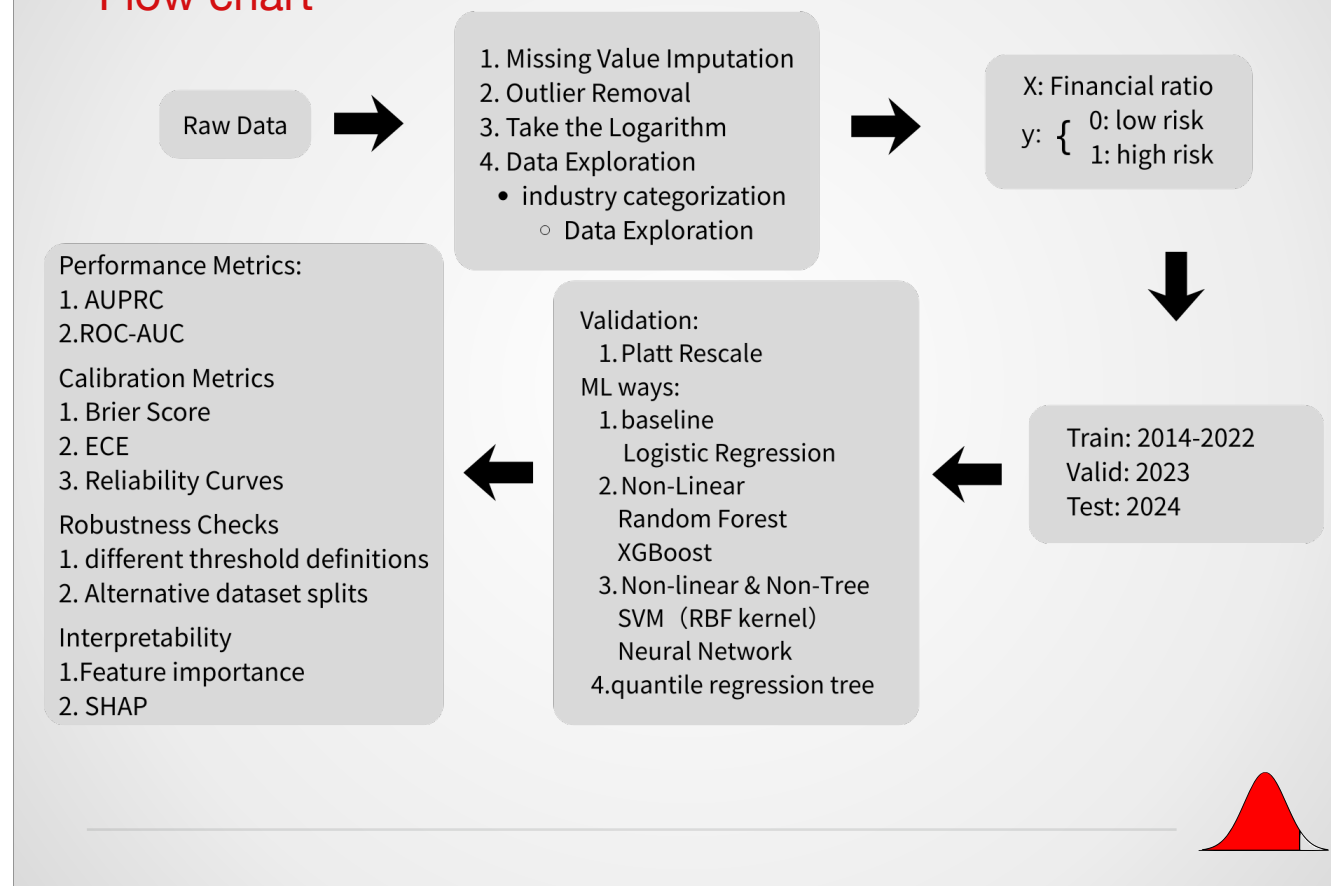
At this operating point, we capture about 82% of truly high-risk firms (Recall 0.821; TP=147, FN=32).

Among firms flagged as high risk, about 60% are truly high risk (Precision 0.612; TP=147, FP=93; total alerts 240).

Overall accuracy is 0.798, though we focus on Precision/Recall/F1 for decision quality.

So we think this baseline model can catch more high-risk cases. If the policy prefers fewer false alarms (higher precision), we can raise the threshold, accepting lower recall—it's a tunable trade-off based on review capacity.

Flow chart



This page summarizes the forecast year 2024 result, we use 2023 financial ratios to predict whether 2024 is high risk or not.

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Results

Model Name	AUC	AUPRC	Brier Score	ECE
Logistic Regression	0.8018	0.6542	0.1575	0.0633
Logistic Regression (no GICS)	0.7920	0.6533	0.1589	0.0681
Random Forest	0.8133	0.6662	0.1503	0.0571
XGBoost	0.8222	0.6864	0.1499	0.0638
SVM	0.7751	0.5780	0.1701	0.0583
Neural Network	0.8154	0.6674	0.1553	0.0634
Quantile Random Forest	0.8173	0.6840	0.1490	0.0715

- ▣ XGBoost has the highest AUC (0.8222) and AUPRC (0.6864)
- ▣ baseline Logistic Regression outperforms the Logistic model without GICS
- ▣ Random Forest achieves the lowest ECE, implying it provides the most well-calibrated probabilities



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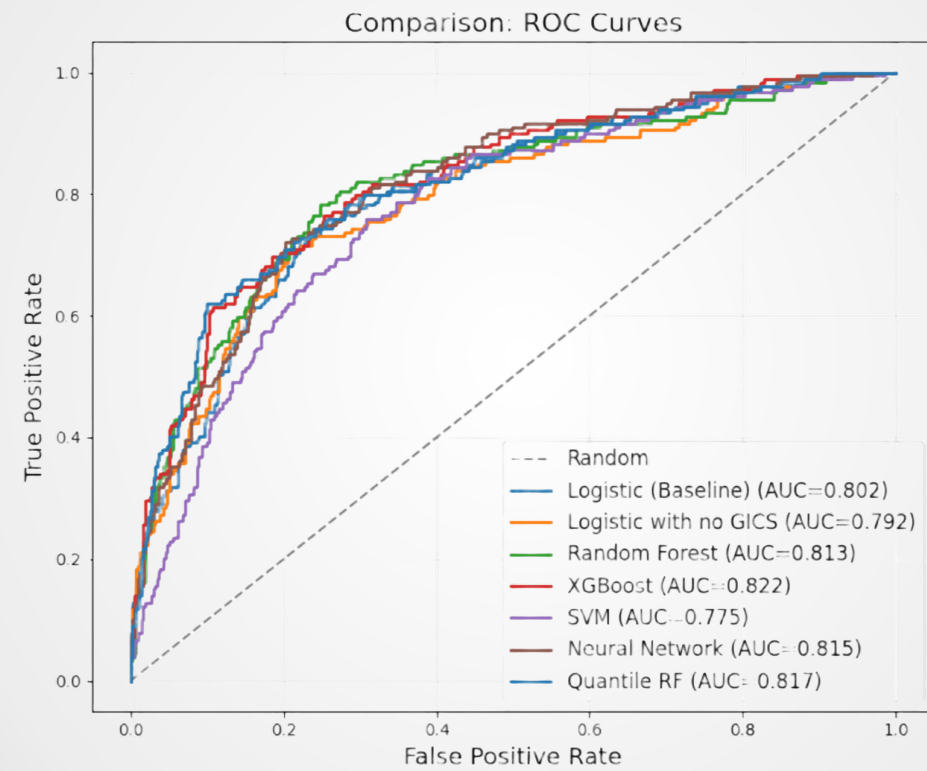
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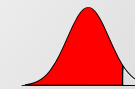
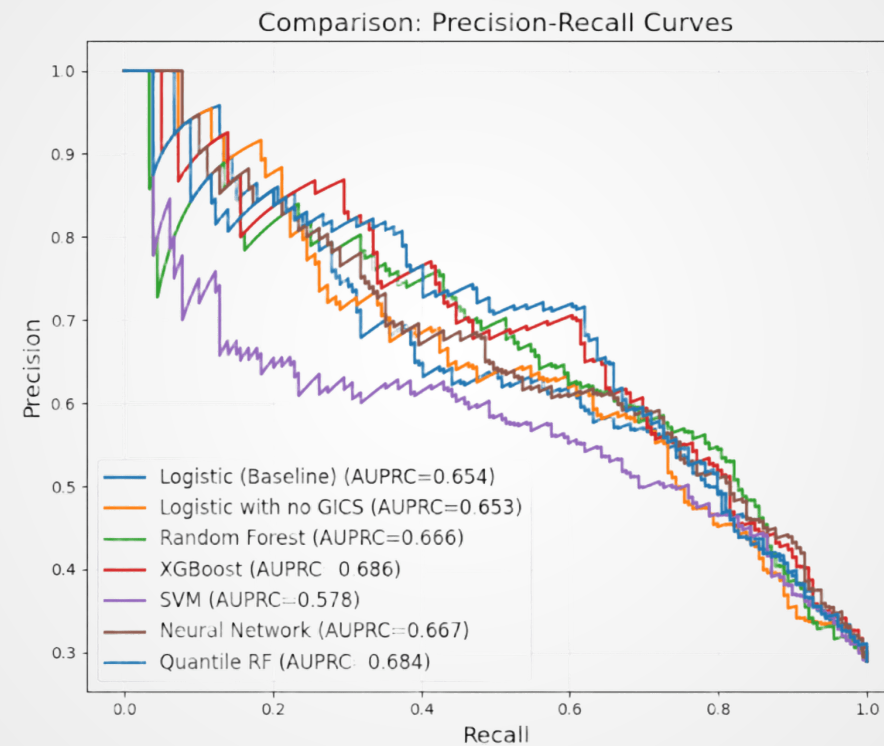


We start with the ROC Curves to see the general performance.

As you can see, almost all our models performed well, beating the random guess line.

- The Red line, which represents XGBoost, achieved the highest AUC of 0.822.
- This shows that XGBoost has the best ability to distinguish between high-risk and low-risk companies. (比benchmark好)

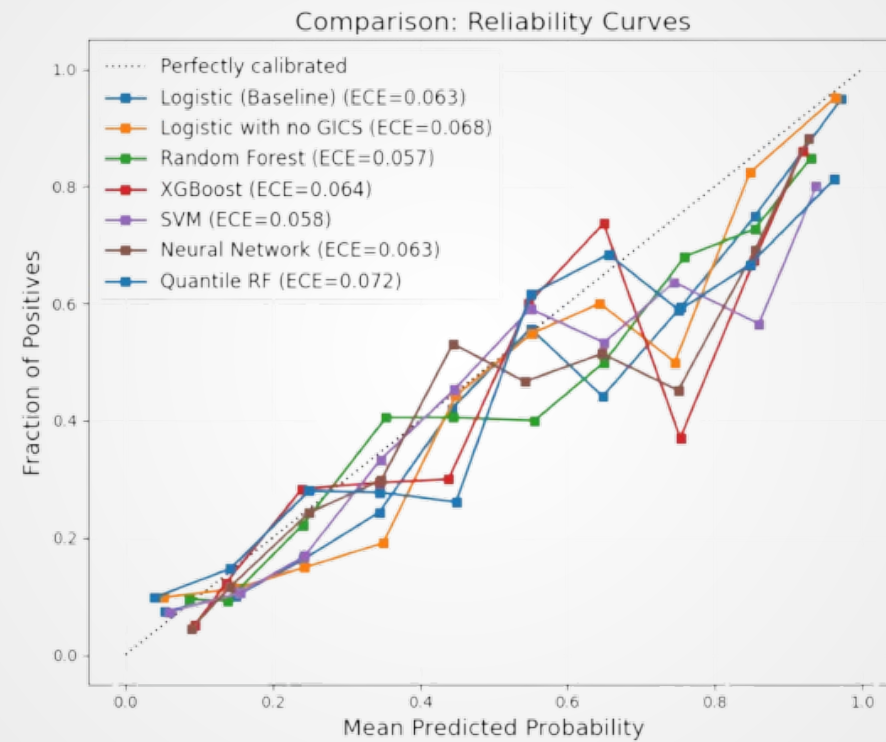
Results



However, since financial defaults are rare, we must also look at the Precision-Recall Curves. This metric is more reliable for imbalanced datasets.

- Looking at this graph, XGBoost (the red line) is still the winner with an AUPRC of 0.686.
- It is followed closely by the Quantile Random Forest (the blue line).
- This confirms that our model is robust even when dealing with rare default events.

Results



Here, the dotted diagonal line represents perfect calibration. The closer to this line, the better.

- You can see that our Random Forest (green line) follows the diagonal line very closely with the lowest ECE error of 0.057.
- This means when our model predicts a risk probability, it is highly accurate and trustworthy for real-world decision-making.

Results - Robustness Test

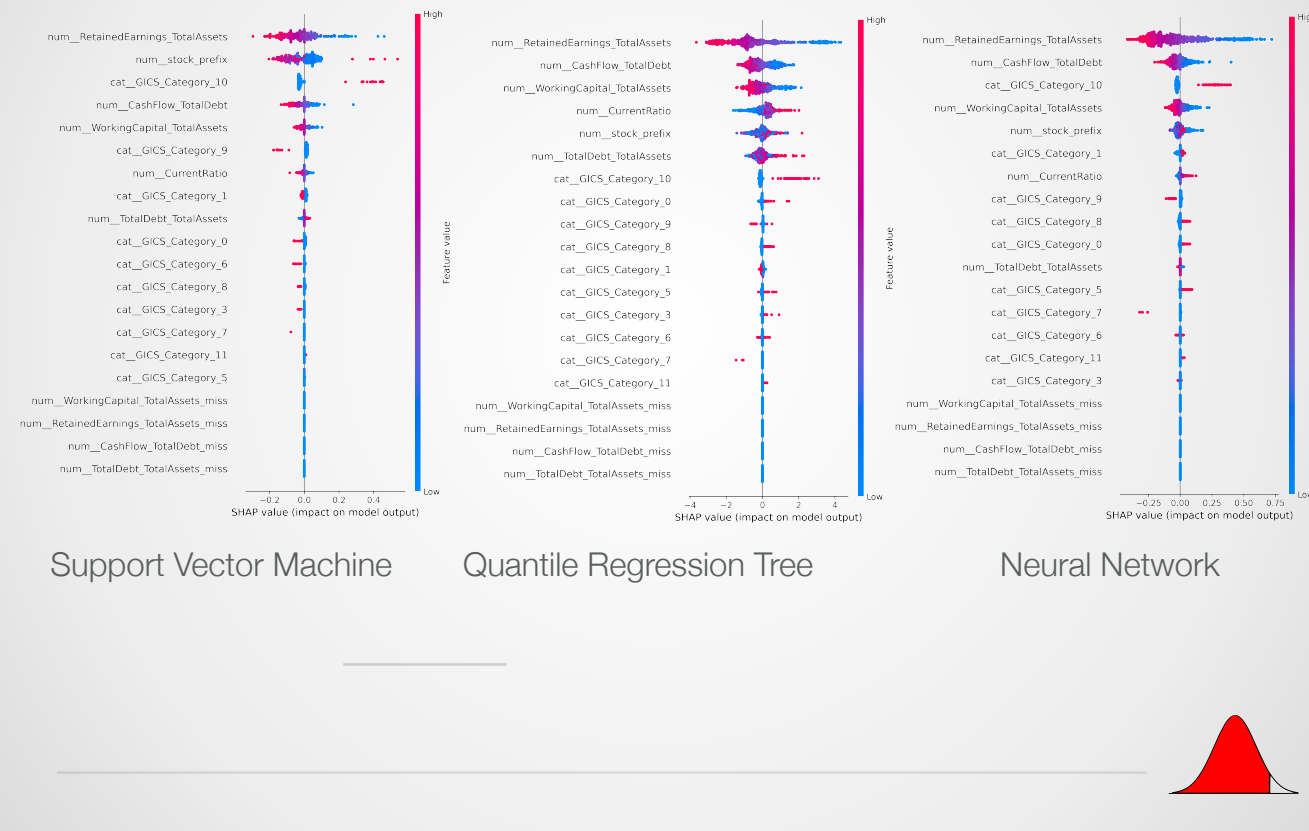
tau	calibration	pr_auc	roc_auc	f1	precision	recall	brier	ece
6	platt 	0.748767	0.696226 	0.729560	0.580000	0.983051	0.235200	0.140743
7	platt	0.654198	0.801778	0.597701	0.615385	0.581006	0.157525	0.063335
8	platt 	0.577760	0.890327 	0.444444	0.750000	0.315789	0.073633	0.031298
6	raw	0.748767	0.696226	0.737819	0.625984	0.898305	0.232563	0.119627
7	raw	0.654198	0.801778	0.612987	0.572816	0.659218	0.169858	0.112107
8	raw	0.577760	0.890327	0.573034	0.500000	0.671053	0.113915	0.166093



We conducted robustness tests using different TCRI thresholds to evaluate financial risk models.

- As we increased the risk threshold from 6 to 8, we observed a clear trade-off. Our ROC-AUC score improved significantly to 0.89. This happened because companies in extreme distress look very different from healthy ones, making them easier to distinguish. However, because these extreme cases are so rare, our PR-AUC naturally declined due to data imbalance.
- The critical finding was the impact of Platt Scaling. While calibration didn't change much at lower thresholds, it was vital at the strictest level. It effectively reduced the Expected Calibration Error (ECE). This confirms that for detecting rare financial risks, probability calibration is a necessary step to ensure reliability.

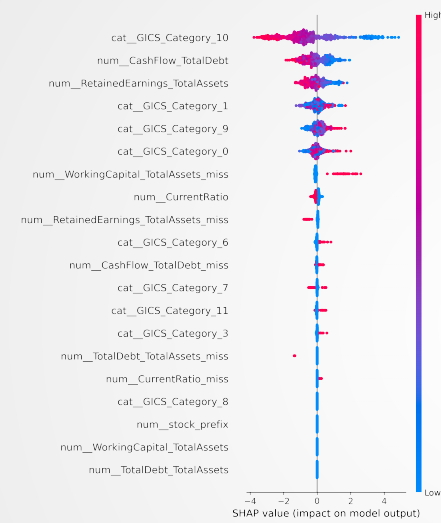
Results SHAP



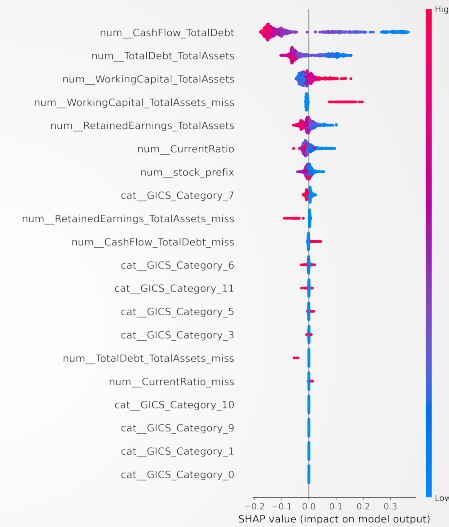
First, let's compare the SVM, QRT, and Neural Network models. We see a similarity among these three.

- They all focused on the 'Retained Earnings to Total Assets' ratio, which measures a company's ability to accumulate profit over time. This finding aligns with the pecking order theory. (The pecking order theory states that firms prefer internal financing first, then debt, and issue equity only as a last resort due to information asymmetry.)

Results SHAP



XGBoost



Random Forest



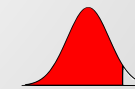
Moving on to XGBoost and Random Forest.

- Unlike the previous models that focused on financial ratios, xgboost dominated by industry sector—specifically, GICS Category 10, which represents Real Estate with tangible nature of asset collateral and less volatility in cash flow structure.
- In contrast, the Random Forest model takes a completely different approach. It focuses on immediate financial health. It prioritizes leverage and solvency metrics—like Cash Flow to Total Debt

Conclusion

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- ▣ XGBoost has best performance
- ▣ Industry is helpful in prediction
- ▣ Probability calibration is necessary when threshold is high.





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